



The objective of this project is to analyse customer retention, loyalty behaviour, churn patterns, promotion effectiveness, and store/channel performance of **TARGET** using Power BI. The dashboard helps identify customer segments and provides actionable recommendations to improve retention and revenue.

TASK- 1 : Data Modeling & Cleaning

ALL MEASURES-

Average CLV = AVERAGEX(ALL(Customer_Demographics),[CLV])

Average Customer CLV = AVERAGE(Customer_Demographics[Customer CLV])

Average Purchase Amount = AVERAGE(Customer_Transactions[Amount])

Average Transaction Value = AVERAGE(Customer_Transactions[Amount])

Avg Purchase Frequency = AVERAGE(Customer_Demographics[Purchase Count])

Avg Transaction Amount = AVERAGE(Customer_Transactions[Amount])

Churn Rate = DIVIDE([Churned Customers],[Total Customers],0)

ChurnedCustomers=CALCULATE(DISTINCTCOUNT(Churn_Labelled_Customers[Customer_ID]),Churn_Labelled_Customers[Churn_Flag] = 1)

CLV = DIVIDE([Total Revenue],AVERAGE(Customer_Demographics[Membership Duration Years]),0)

Promotion % = DIVIDE([Promotion Transactions],[Total Transactions],0)

Promotion Transactions =
CALCULATE(COUNTROWS(Customer_Transactions),Customer_Transactions[Promotion_Applied] = "Yes")

Repeat Customers =
CALCULATE(DISTINCTCOUNT(Customer_Transactions[Customer_ID]),FILTER(

VALUES(Customer_Transactions[Customer_ID]),CALCULATE(COUNT(Customer_Transactions[Transaction_ID])) > 1))

Repeat rate = DIVIDE([Repeat Customers],[Total Customers],0)

Retained Customers = [Total Customers] - [Churned Customers]

Retention rate = DIVIDE([Retained Customers],[Total Customers],0)

Total Customers = DISTINCTCOUNT(Customer_Demographics[Customer_ID])

Total Revenue = SUM(Customer_Transactions[Amount])

Total Transactions = COUNT(Customer_Transactions[Transaction_ID])

High CLV % =

DIVIDE(CALCULATE(DISTINCTCOUNT(Customer_Demographics[Customer_ID]),Customer_Demographics[CLV Segment] = "High CLV"),DISTINCTCOUNT(Customer_Demographics[Customer_ID]),0)

Target_Retention

File Home Transform Add Column View Tools Help

Close & Apply New Source Recent Enter Data Data source settings Manage Parameters Refresh Preview Manage Query Choose Columns Remove Columns Keep Rows Remove Rows Sort Split Column Group By Data Type: Text Use First Row as Headers Replace Values Merge Queries Append Queries Combine Files Combine

Queries [7]

Customer_Demographics

Customer_Transactions

Store_Locations

Loyalty_Program

Churn_Labelled_Custom...

Measure_Table

Merged_Transactions_St...

Table.TransformColumnTypes(*Added Custom*,({"Membership_Duration", Int64.Type), ("Customer_ID", type text), ("Age", Int64.Type)})

Customer_ID Age Gender Region Income_Level Membership_Slice Preferred_C

Valid 100% Valid 100% Valid 100% Valid 100% Valid 100% Valid 100% Valid 100% Valid 100% Valid 100%

Error 0% Error 0% Error 0% Error 0% Error 0% Error 0% Error 0% Error 0% Error 0%

Empty 0% Empty 0% Empty 0% Empty 0% Empty 0% Empty 0% Empty 0% Empty 0%

1 C001 58 Female North Low 18-08-2023 Online

2 C002 48 Male North High 17-09-2023 Store

3 C003 34 Female East Medium 06-09-2023 Store

4 C004 27 Female Central Low 30-06-2022 Store

5 C005 40 Male South High 12-09-2022 Online

6 C006 58 Male South High 30-10-2018 Online

7 C007 38 Female East High 01-10-2019 Store

8 C008 42 Female South Low 25-02-2018 Online

9 C009 30 Male North Medium 27-12-2019 Online

10 C010 30 Female Central Medium 10-06-2019 Online

11 C011 43 Male West Medium 19-04-2019 Store

12 C012 55 Female South High 28-01-2018 Store

13 C013 59 Male North Medium 27-10-2023 Online

14 C014 43 Male West High 31-12-2018 Store

15 C015 22 Male Central High 10-06-2019 Online

16 C016 41 Female West High 28-08-2021 Online

17 C017 21 Female North Medium 01-06-2020 Online

18 C018 43 Male West Low 27-10-2020 Online

19 C019 49 Female East Medium 23-10-2020 Store

20 C020 57 Male West High 05-07-2022 Store

21 C021 21 Male South Low 07-07-2020 Store

22 C022 40 Female South Medium 07-08-2024 Online

23 C023 52 Female East Low 10-11-2022 Online

24 C024 31 Male North Medium 12-09-2022 Online

25 C025 42 Female South High 22-09-2023 Store

26

9 COLUMNS, 200 ROWS Column profiling based on top 1000 rows

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Target_Retention

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Queries [7]

Customer_Demographics

Customer_Transactions

Store_Locations

Loyalty_Program

Churn_Labelled_Custom...

Measure_Table

Merged_Transactions_St...

Table.RenameColumns(*Inserted Month Name*,({"Year", "Transaction_Year"}, {"Month Name", "Transaction_Month"}))

Transaction_ID Customer_ID Store_ID Product_Category Transaction_Date Amount Promotion

Valid 100% Valid 100% Valid 100% Valid 100% Valid 100% Valid 100% Valid 100% Valid 100% Valid 100%

Error 0% Error 0% Error 0% Error 0% Error 0% Error 0% Error 0% Error 0% Error 0%

Empty 0% Empty 0% Empty 0% Empty 0% Empty 0% Empty 0% Empty 0% Empty 0%

1 T0001 C051 S17 Home & Living 30-12-2023 128.29 No

2 T0002 C072 S06 Home & Living 23-09-2024 788.59 No

3 T0003 C110 S08 Apparel 25-04-2023 419.72 No

4 T0004 C044 S19 Groceries 25-08-2023 333.89 Yes

5 T0005 C087 S19 Home & Living 05-10-2023 671.25 No

6 T0006 C014 S14 Groceries 17-03-2024 722.45 No

7 T0007 C197 S13 Groceries 16-03-2023 661.62 No

8 T0008 C115 S03 Home & Living 11-02-2023 469.68 Yes

9 T0009 C197 S05 Home & Living 17-01-2023 336.09 No

10 T0010 C050 S16 Home & Living 10-08-2024 181.01 Yes

11 T0011 C092 S03 Electronics 27-01-2023 722.01 No

12 T0012 C136 S17 Apparel 17-06-2023 215.57 No

13 T0013 C121 S08 Groceries 30-03-2024 467.17 No

14 T0014 C149 S13 Home & Living 20-04-2024 621.31 No

15 T0015 C021 S20 Apparel 10-05-2023 149.7 No

16 T0016 C053 S04 Apparel 19-10-2023 621.37 Yes

17 T0017 C097 S05 Groceries 23-10-2023 361.01 No

18 T0018 C046 S06 Electronics 06-07-2023 251.75 No

19 T0019 C196 S03 Home & Living 08-04-2023 329.39 No

20 T0020 C131 S01 Electronics 10-02-2024 591.41 Yes

21 T0021 C068 S12 Home & Living 01-07-2024 441.08 Yes

22 T0022 C021 S03 Electronics 01-09-2023 60.11 No

23 T0023 C005 S09 Groceries 31-12-2023 140.64 No

24 T0024 C002 S14 Groceries 25-03-2024 103.15 Yes

25 T0025 C120 S03 Groceries 17-01-2023 341.98 No

26

9 COLUMNS, 999+ ROWS Column profiling based on top 1000 rows

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Target_Retention

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Queries [7]

- Customer_Demographics
- Customer_Transactions
- Store_Locations
- Loyalty_Program
- Churn_Labelled_Custom...
- Measure_Table
- Merged_Transactions_St...

Table.TransformColumnTypes(*Removed Duplicates*,({Store_ID, type text}, {"Store_Type", type text}, {"Region", type text}, {"Opening_Year", type text}))

	Store_ID	Store_Type	Region	Opening_Year
1	501	Online	East	2018
2	502	Neighborhood Market	West	2017
3	503	Online	West	2009
4	504	Online	Central	2010
5	505	Online	Central	2014
6	506	Online	South	2014
7	507	Superstore	South	2019
8	508	Neighborhood Market	North	2011
9	509	Superstore	Central	2012
10	510	Superstore	East	2012
11	511	Neighborhood Market	Central	2013
12	512	Neighborhood Market	East	2021
13	513	Superstore	West	2016
14	514	Neighborhood Market	Central	2009
15	515	Online	West	2012
16	516	Neighborhood Market	South	2019
17	517	Neighborhood Market	South	2010
18	518	Online	North	2011
19	519	Online	Central	2017
20	520	Neighborhood Market	West	2013

4 COLUMNS, 20 ROWS Column profiling based on top 1000 rows

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Target_Retention

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Close & Apply New Recent Enter Data Data source settings Manage Parameters Refresh Preview Manage Query Choose Columns Remove Columns Keep Rows Remove Rows Sort Split Column Group By Data Type: Text Use First Row as Headers Replace Values Merge Queries Append Queries Combine Files Combine

Queries [7]

- Customer_Demographics
- Customer_Transactions
- Store_Locations
- Loyalty_Program
- Churn_Labelled_Custom...
- Measure_Table
- Merged_Transactions_St...

Table.TransformColumnTypes(*Removed Duplicates*,({Points_Redemed, Int64.Type}, {"Points_Earned, Int64.Type}, {"Loyalty_Tier", type text}))

	Customer_ID	Loyalty_Tier	Points_Earned	Points_Redemed
1	C001	Elite	3373	216
2	C002	Premium	2651	1286
3	C003	Premium	2945	452
4	C004	Basic	2456	664
5	C005	Plus	3427	575
6	C006	Plus	962	2027
7	C007	Premium	2274	3228
8	C008	Elite	2545	719
9	C009	Basic	1307	2090
10	C010	Elite	686	289
11	C011	Premium	2243	919
12	C012	Premium	1231	3051
13	C013	Basic	2435	3043
14	C014	Basic	1713	829
15	C015	Plus	1921	2084
16	C016	Premium	928	3446
17	C017	Basic	3405	275
18	C018	Plus	1807	1125
19	C019	Elite	2784	2647
20	C020	Premium	1717	2005
21	C021	Elite	1810	1169
22	C022	Plus	1270	1294
23	C023	Plus	1257	2228
24	C024	Premium	635	3451
25	C025	Plus	1291	3395
26	C026	Basic	670	1604

4 COLUMNS, 200 ROWS Column profiling based on top 1000 rows

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Target_Retention

File Home Transform Add Column View Tools Help

Close & Apply New Source Recent Enter Data Data source settings Manage Parameters Refresh Preview Manage Query Choose Columns Remove Columns Keep Rows Remove Rows Sort Split Column Group By Data Type: Text * Use First Row as Headers * Merge Queries * Append Queries * Combine Files * Combine

Queries [7]

Customer_Demographics Customer_Transactions Store_Locations Loyalty_Program Churn_Labelled_Custom... Measure_Table Merged_Transactions_St...

Table.TransformColumnTypes(*Removed Duplicates*({{"Customer_ID", type text}, {"Last_Purchase_Date", type date}, {"Churn_Flag", type text}, {"Churn_Reason", type text}}))

Query Settings

PROPERTIES

Name Churn_Labelled_Customers

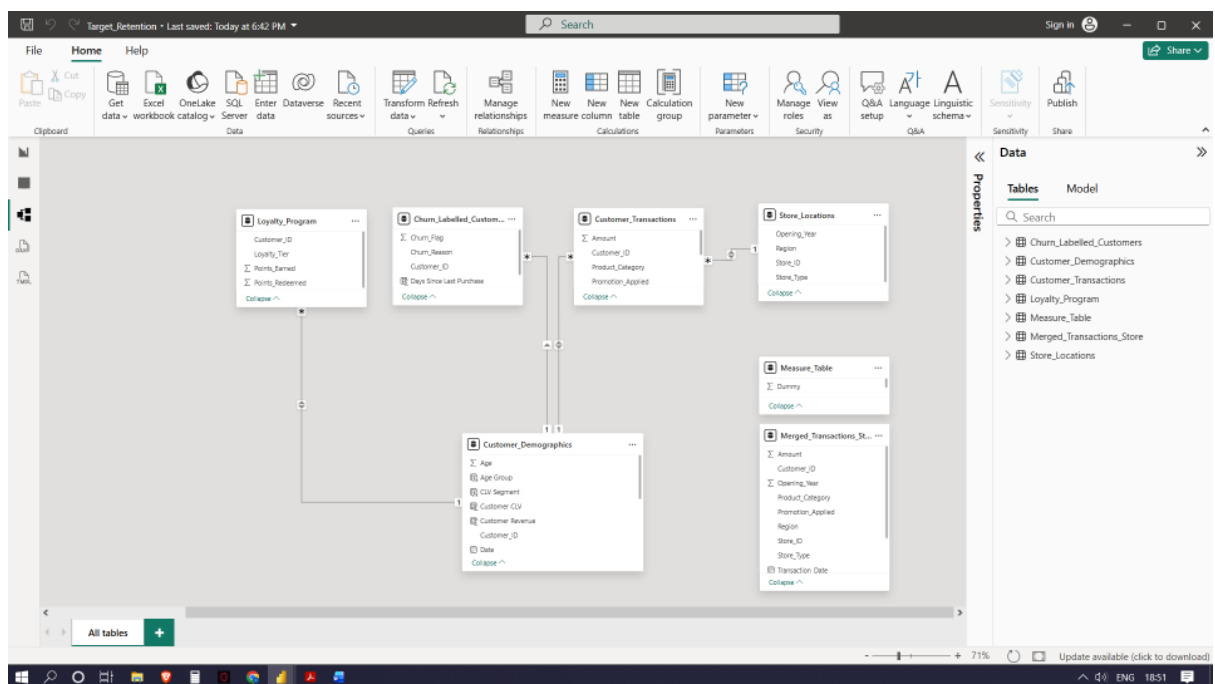
APPLIED STEPS

Source Promoted Headers Changed Type Removed Duplicates Changed Type1

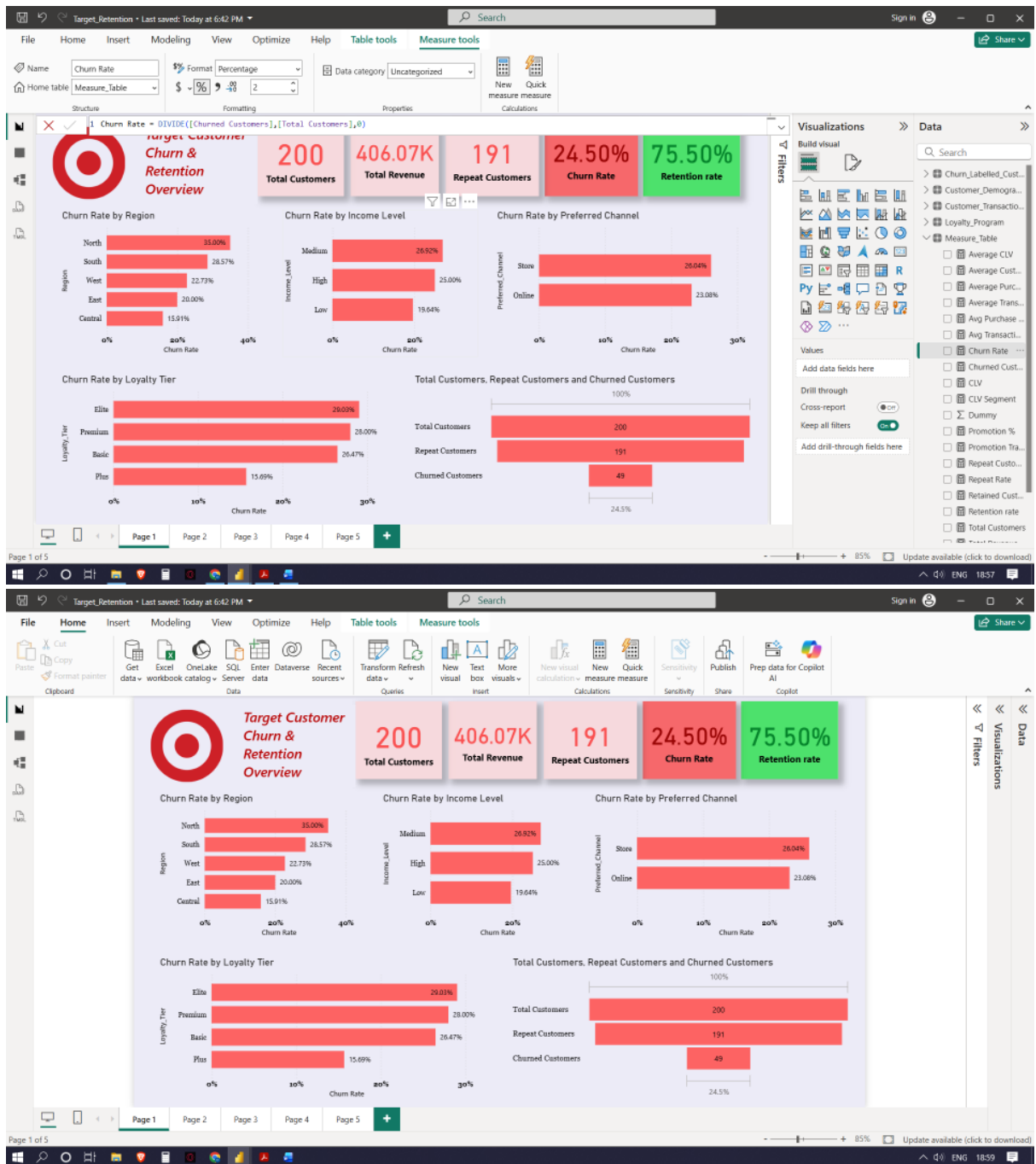
	Customer_ID	Last_Purchase_Date	Churn_Flag	Churn_Reason
1	C001	27-09-2024	0 Active	
2	C002	12-02-2024	0 Inactivity	
3	C003	21-01-2024	0 Low Engagement	
4	C004	22-07-2024	0 Active	
5	C005	19-03-2023	0 Inactivity	
6	C006	13-07-2023	0 Active	
7	C007	12-10-2023	1 Active	
8	C008	10-06-2024	0 Active	
9	C009	07-06-2024	0 Active	
10	C010	04-09-2024	0 Active	
11	C011	05-06-2023	0 Active	
12	C012	09-12-2023	0 Inactivity	
13	C013	05-05-2023	0 Inactivity	
14	C014	07-10-2023	1 Active	
15	C015	13-09-2024	0 Low Engagement	
16	C016	19-10-2024	0 Active	
17	C017	08-05-2023	0 Active	
18	C018	10-11-2023	0 Active	
19	C019	28-11-2023	0 Active	
20	C020	30-05-2023	0 Active	
21	C021	29-03-2024	0 Active	
22	C022	05-03-2023	0 Active	
23	C023	24-02-2024	0 Competitor	
24	C024	22-01-2023	0 Active	
25	C025	29-07-2023	0 Active	
26	C026	30-09-2024	0 Active	

4 COLUMNS, 260 ROWS Column profiling based on top 1000 rows

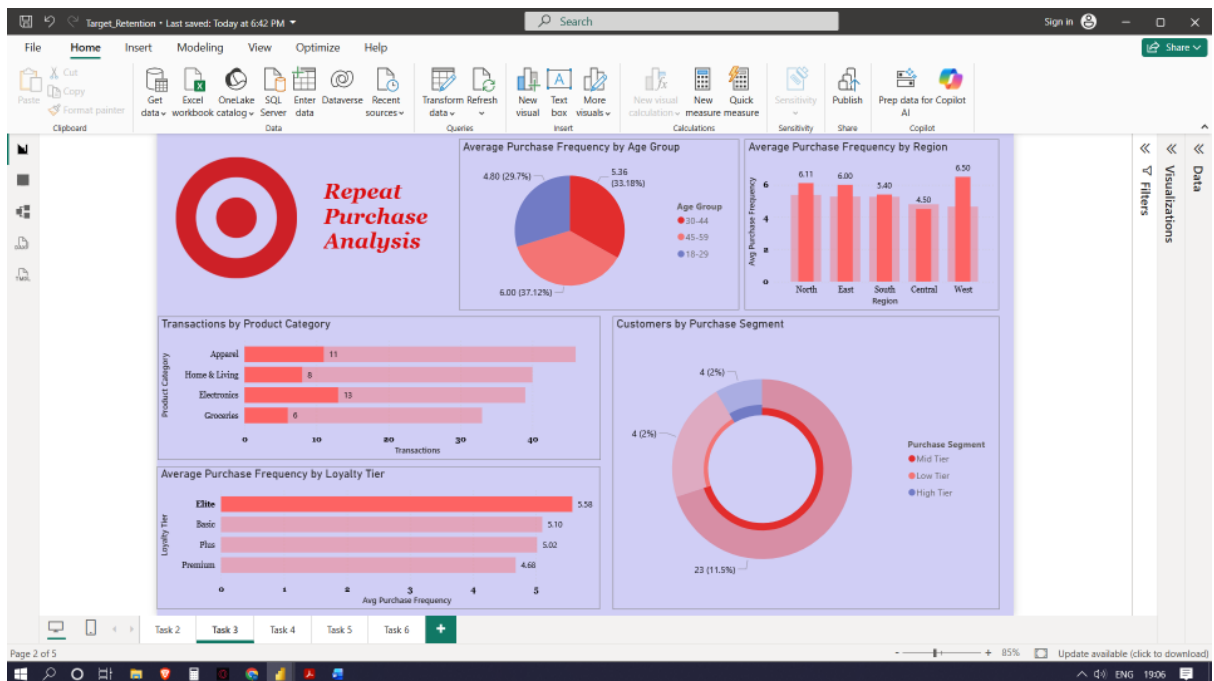
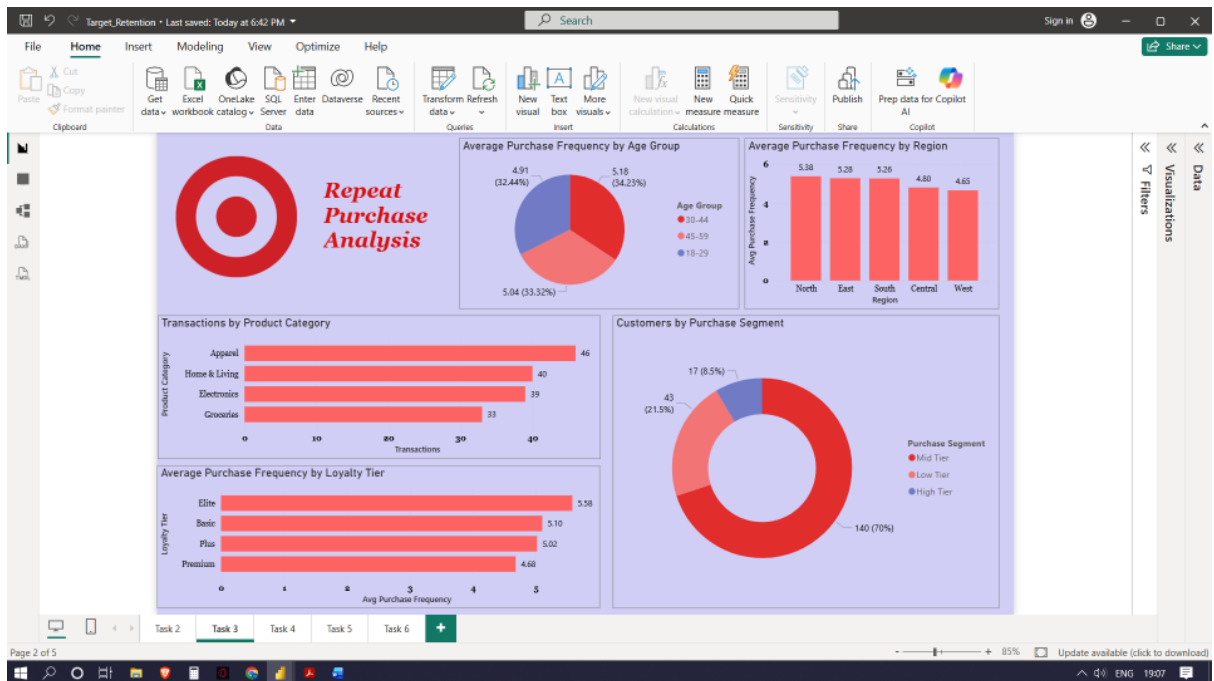
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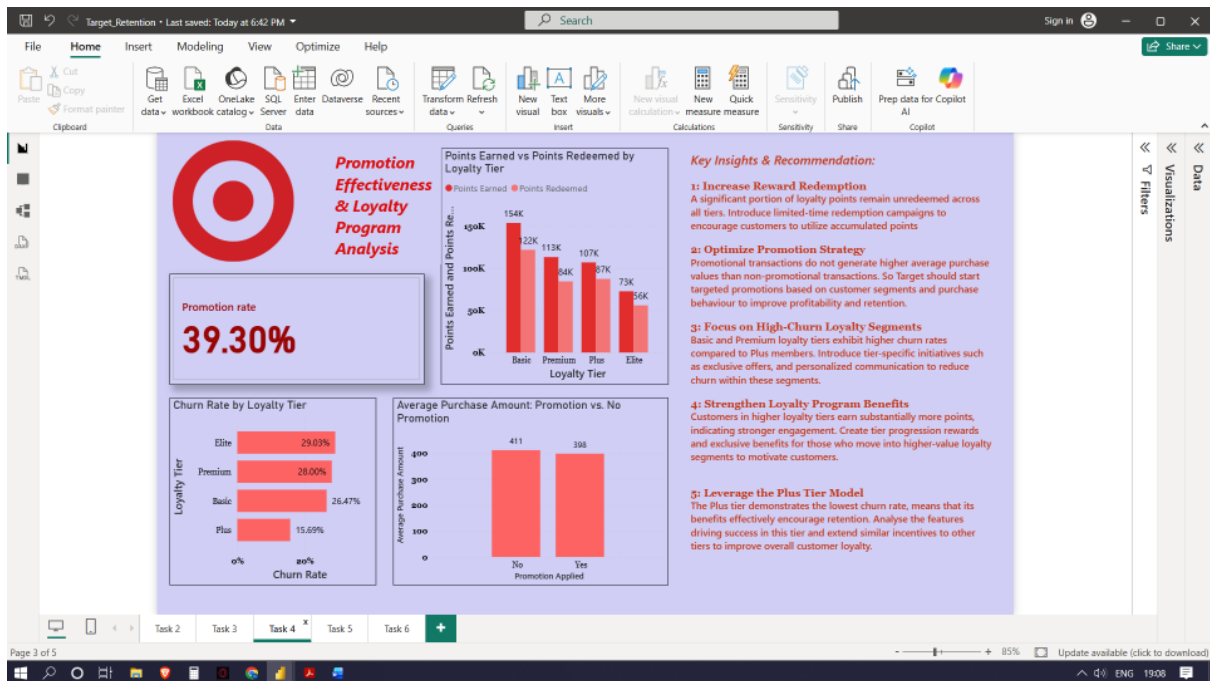
TASK-2: Churn & Retention Metrics



TASK-3: Repeat Purchase Analysis



Task 4: Promotion & Loyalty Impact



Task 5: Store & Channel Performance vs Retention

Merged store data with Transactions

Target_Retention • Last saved: Today at 6:42 PM

Search

File Home Help Table tools

Get data Excel OneLake SQL Server Data Transform Refresh data Queries Manage relationships New measure measure column Calculations New table New table Manage roles View as Sensitivity Publish

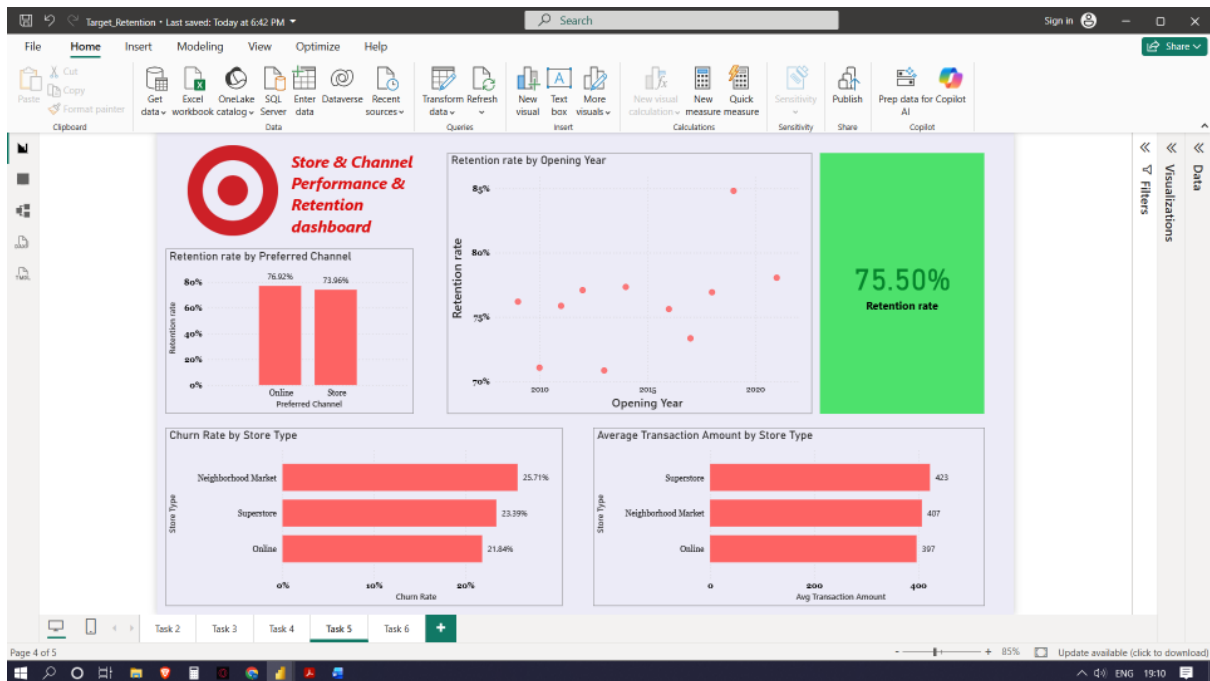
Clipboard

Data

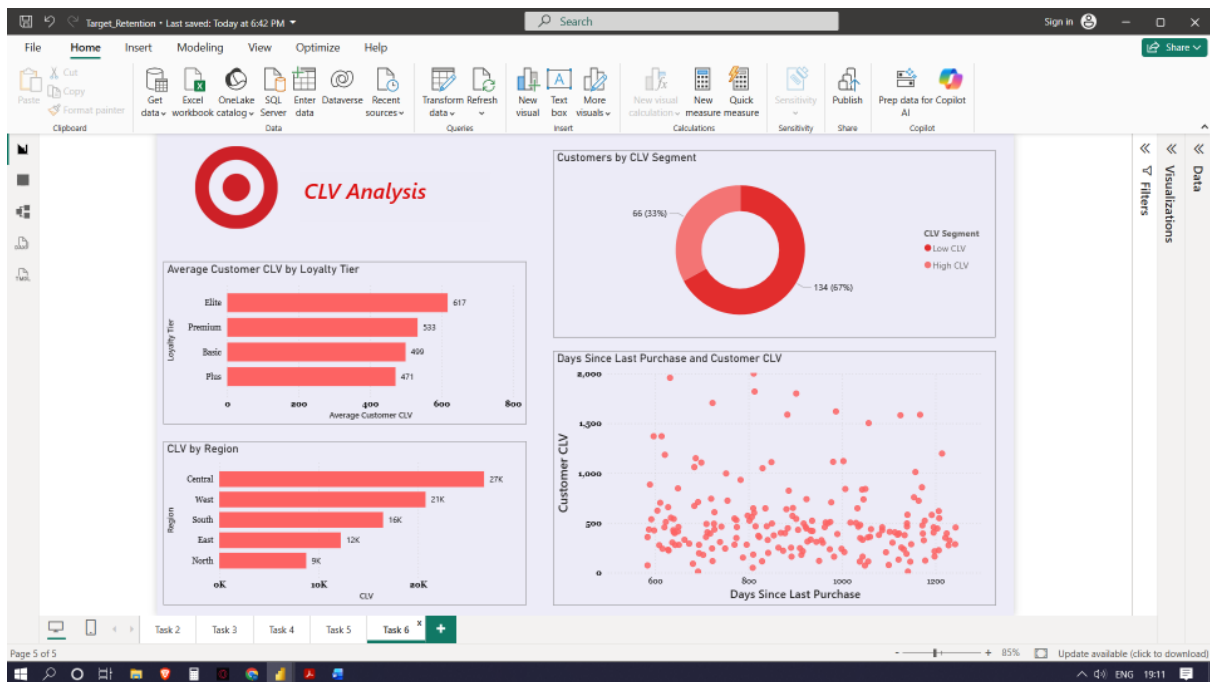
Transaction_ID	Customer_ID	Store_ID	Product_Category	Transaction_Date	Amount	Promotion_Applied	Transaction_Year	Transaction_Month	Store_Type	Region	Opening
T0004	C044	S19	Groceries	25 August 2023	332.99	Yes	2023	August	Online	Central	
T0005	C087	S19	Home & Living	05 October 2023	671.25	No	2023	October	Online	Central	
T0016	C053	S04	Apparel	19 October 2023	621.37	Yes	2023	October	Online	Central	
T0009	C197	S05	Home & Living	17 January 2023	336.09	No	2023	January	Online	Central	
T0017	C097	S05	Groceries	23 October 2023	361.01	No	2023	October	Online	Central	
T0031	C012	S04	Electronics	12 January 2023	442.68	No	2023	January	Online	Central	
T0033	C141	S19	Apparel	19 July 2023	214.74	No	2023	July	Online	Central	
T0050	C023	S05	Electronics	01 October 2023	752.47	No	2023	October	Online	Central	
T0055	C101	S19	Electronics	06 December 2023	595.31	Yes	2023	December	Online	Central	
T0056	C167	S19	Apparel	03 January 2023	384.5	Yes	2023	January	Online	Central	
T0060	C028	S19	Home & Living	20 May 2023	766	No	2023	May	Online	Central	
T0064	C166	S05	Groceries	24 July 2024	333.23	No	2024	July	Online	Central	
T0071	C061	S04	Groceries	25 May 2023	140.04	No	2023	May	Online	Central	
T0093	C184	S04	Home & Living	15 February 2023	274.62	No	2023	February	Online	Central	
T0125	C076	S04	Groceries	15 May 2024	229.81	No	2024	May	Online	Central	
T0130	C130	S19	Groceries	29 February 2024	50.9	No	2024	February	Online	Central	
T0134	C100	S19	Electronics	30 August 2023	274.64	No	2023	August	Online	Central	
T0136	C011	S19	Apparel	13 July 2023	782.42	Yes	2023	July	Online	Central	
T0137	C127	S19	Home & Living	02 May 2023	644.46	Yes	2023	May	Online	Central	
T0140	C119	S19	Apparel	25 December 2022	498.1	Yes	2023	December	Online	Central	
T0143	C153	S04	Groceries	07 January 2023	535.44	No	2023	January	Online	Central	
T0144	C049	S05	Apparel	25 January 2024	657.49	No	2024	January	Online	Central	
T0145	C172	S19	Apparel	29 May 2024	51.79	Yes	2024	May	Online	Central	
T0156	C027	S19	Apparel	08 June 2023	344.41	No	2023	June	Online	Central	
T0163	C063	S05	Home & Living	13 June 2024	512.62	No	2024	June	Online	Central	
T0172	C033	S05	Apparel	28 October 2023	204.88	Yes	2023	October	Online	Central	
T0179	C139	S04	Apparel	09 July 2024	368.17	No	2024	July	Online	Central	
T0183	C109	S04	Electronics	04 February 2023	372.72	No	2023	February	Online	Central	

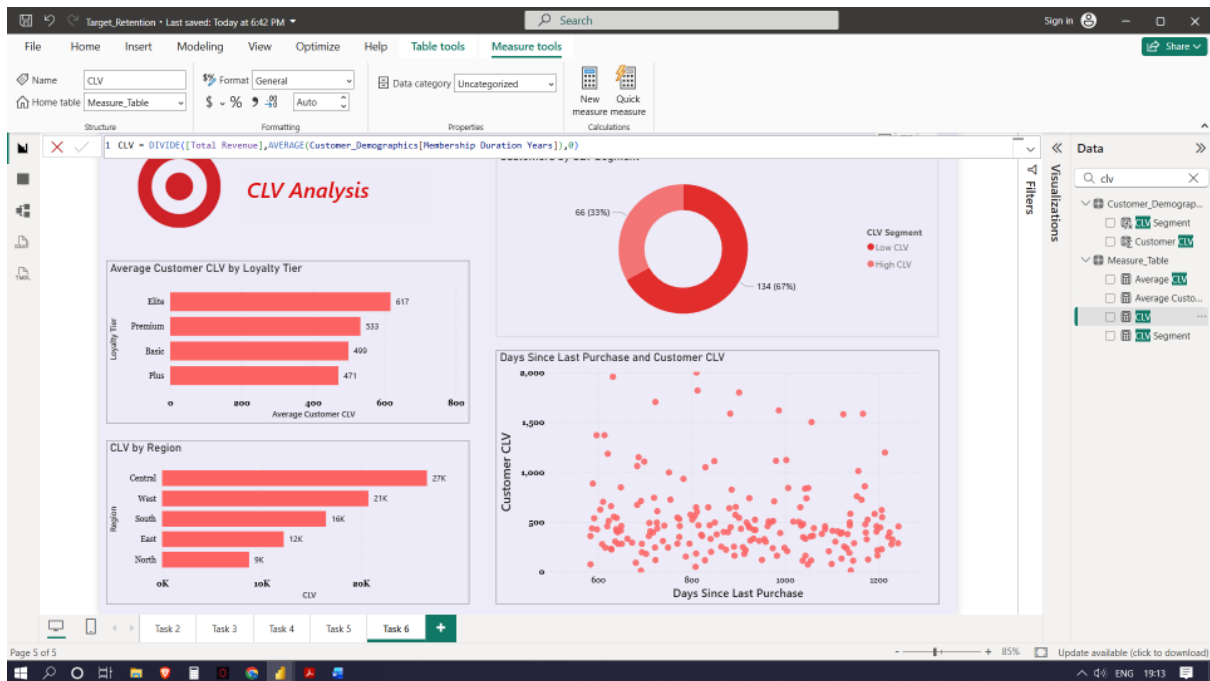
Table: Merged_Transactions_Store (1,000 rows)

Update available (click to download)



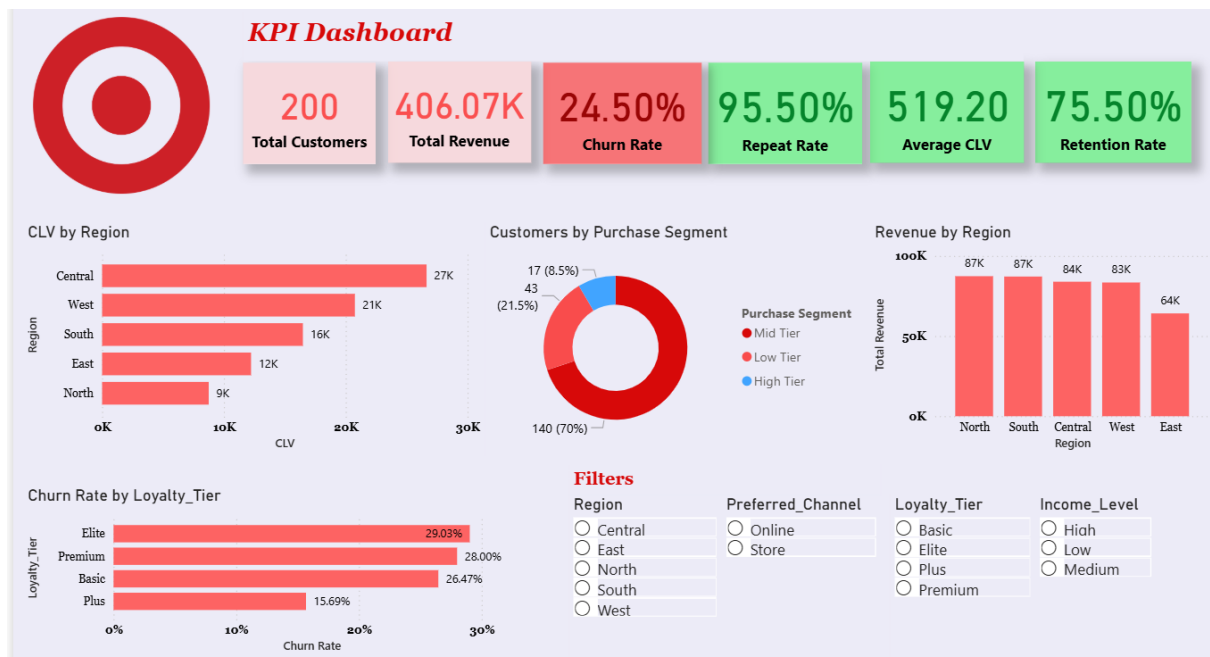
Task 6: Customer Lifetime Value (CLV) Analysis



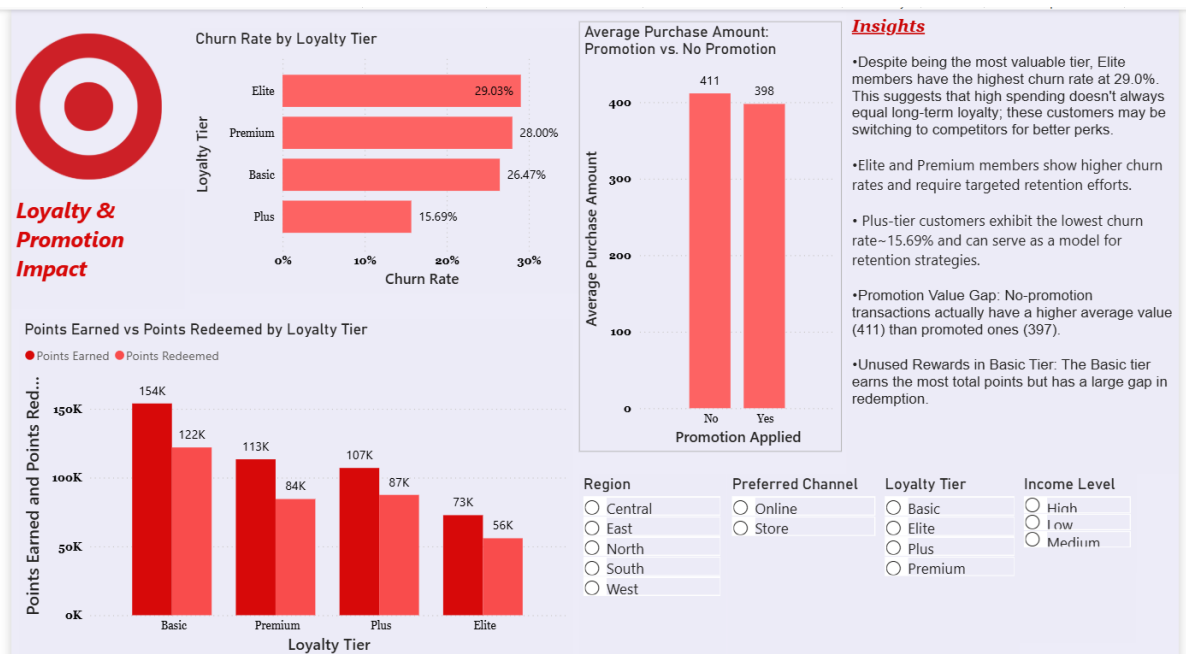


Task 7: Final Dashboard & Executive Summary

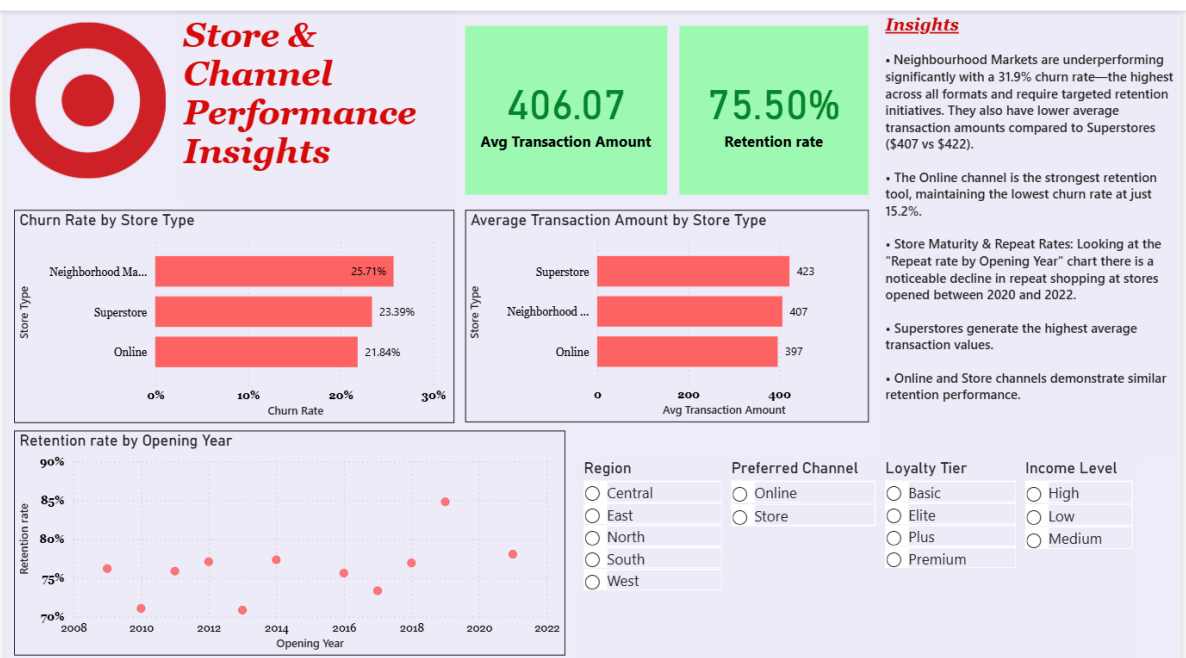
Page 1-



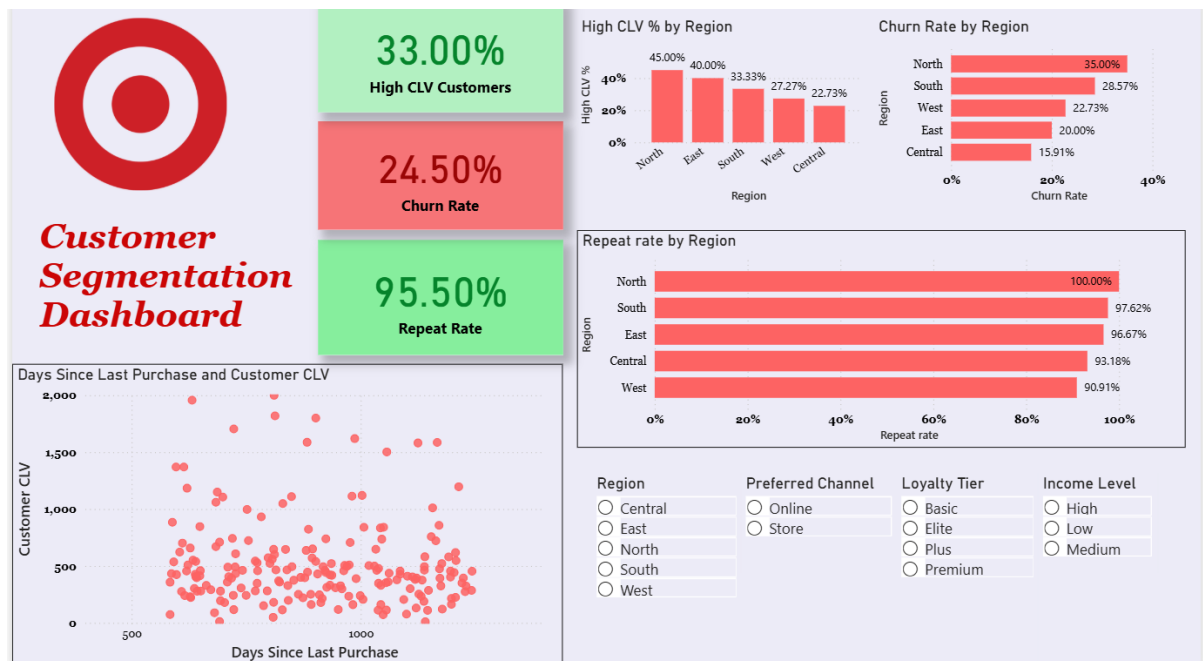
Page 2-



Page 3-



Page 4-



Recommendations-



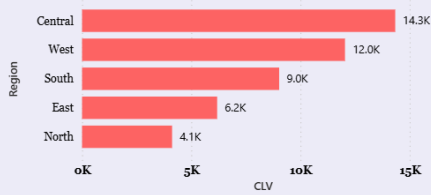
Filters applied and in sync-



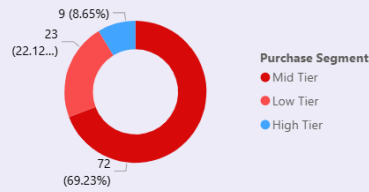
KPI Dashboard

104	216.49K	23.08%	96.15%	540.22	76.92%
Total Customers	Total Revenue	Churn Rate	Repeat Rate	Average CLV	Retention Rate

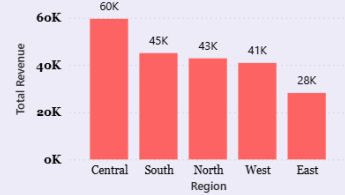
CLV by Region



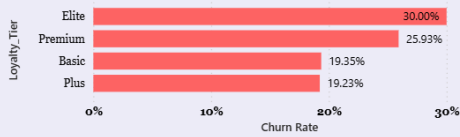
Customers by Purchase Segment



Revenue by Region



Churn Rate by Loyalty_Tier



Filters

Region

- ☐ Central
- ☐ East
- ☐ North
- ☐ South
- ☐ West

Preferred_Channel

- ☒ Online
- ☐ Store

Loyalty_Tier

- ☐ Basic
- ☐ Elite
- ☐ Plus
- ☐ Premium

Income_Level

- ☐ High
- ☐ Low
- ☐ Medium